

Roljohn Frilles

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Co-Curricular Experience

AWS CLOUD CLUB Skill Building - PUP MANILA

Manila, Philippines

ML/AI Co-Head

2023 - 2024

- Led a hands-on workshop introducing foundational Python concepts essential for AI and machine learning, covering key libraries such as NumPy, Matplotlib, and Pandas.

Education

POLYTECHNIC UNIVERSITY OF THE PHILIPPINES - MANILA

Manila, Philippines

Bachelor of Science in Computer Science

October 2022 - Present

- Consistent President's Lister.
- DOST-SEI Scholar: (2022 – Present)

NEW ERA UNIVERSITY

Quezon City, Philippines

Science, Technology, Engineering, Mathematics Strand

2020 – 2022

- With Academic Honors

Leadership & Activities

DEVCON Manila Chapter

Manila, Philippines

Technical Speaker (Push2Prod 2026)

June 2026

- Delivered a 25-minute technical presentation titled "Preparing for the Real World: Testing, Improving, and Deploying Your Project" to an audience of over 150 IT/CS students.
- Showcased the end-to-end deployment pipeline of a machine learning architecture, translating local Jupyter notebook models into production-ready cloud microservices.

AWS Cloud Club Skill Building

Manila, Philippines

PUP Manila ML/AI Co-Head

2023 - 2024

- Led a hands-on workshop introducing foundational Python concepts essential for AI and machine learning, covering key libraries such as NumPy, Matplotlib, and Pandas.

COLLABOTHON: IDEA JAM

Manila, Philippines

Co-Facilitator

February 2024

- Co-facilitated during the Collabothon: IdeaJam, guiding participants through brainstorming sessions and facilitating discussions on solutions using emerging technologies.

AMAZON WEB SERVICES (AWS) STUDENT COMMUNITY DAY 2024

Manila, Philippines

Technical Support Member

April 2024

- Contributed to the technical execution of the event, ensuring smooth facilitation of sessions and supporting the learning experience for attendees.

Skills

Technical Skills

- Programming Language: Python, HTML, CSS, JavaScript, Flutter
- Data Science & ML: PyTorch, NumPy, Pandas, Matplotlib, Neural Networks, Regression, Classification, Spatio-Temporal Models (STGAE), Graph Attention Networks (GAT)
- Backend & Deployment Tools: FastAPI, Docker, Render, Gradio, Hugging Face, Arduino Uno

Certifications & Awards

Computer Vision: Deep Learning with PyTorch: Object Localization	Coursera Project Network 2023
Computer Vision: Deep Learning with PyTorch: Image Segmentation	Coursera Project Network 2023
Certificate of Appreciation: Collabothon: IdeaJam Co-Facilitator	Department of Budget and Management 2023
AWS Cloud Club Skill Building ML/AI CO-HEAD	Manila, Philippines 2024

Publications

Tutorials Dojo

Technical Author

June 2026

- Authored and published "[My First Exposure to Machine Learning with PyTorch](#)", detailing practical entry points and hands-on experiences with the PyTorch framework.

Projects

HydroCast: Marikina River Basin Flood Forecasting 2026

- Engineered a hybrid deep learning architecture (STGAE, GAT, Transformer) to forecast river water levels up to 48 hours ahead using 13 years of hydrometeorological data.
- Handled 68.9% missing real-world sensor data by implementing unsupervised pre-training for signal imputation.
- Deployed the inference engine as a cloud-ready backend microservice using FastAPI and Docker to decouple ML processing from the mobile frontend.

Kuwartrack: Personal Finance Manager 2025

- Led a team in developing a comprehensive financial tracking mobile application with an intuitive Home dashboard for weekly expense categorization, a Transaction module for budget setting and expense tracking, and detailed Category Breakdown functionality.

Smart Logistics Optimizer (Knapsack Algorithm)

2024

- Created a mobile logistics application that optimizes truck loading efficiency using the knapsack algorithm, intelligently determining optimal item combinations for pallet loading to maximize space utilization and profit margins.

YOLOv8 Object Detection Application

2023

- Developed an interactive machine learning application using Gradio and Hugging Face, leveraging YOLOv8 for precise object detection in uploaded images, providing visual feedback with bounding boxes and classification.

CyberNix-12 Alphabot: Healthcare Companion for Isolated COVID-19 Patients

2020

- Led programming and hardware integration of a healthcare companion robot using Arduino Uno, integrating heart rate, oximeter, and temperature sensors to display vital signs and motivational messages via an LCD interface, and included medication reminders and a mobile UV ste

